

Geometrical Optics

Light as rays: reflection, refraction, fibre optics, mirrors, lenses, and the optics of the human eye.

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When light meets objects much larger than its wavelength (about half a micrometre), we can ignore its wave nature and treat it as **rays**: straight lines along which energy travels. This “geometrical optics” explains mirrors, lenses, prisms and optical fibres, and it is the working language of optometrists, ophthalmologists, camera designers and microscope builders. This chapter develops the laws of reflection and refraction, total internal reflection and fibre optics, image formation by mirrors and lenses, and the optics of the eye and of optical instruments.

YOU WILL BE ABLE TO

- Use the refractive index and Snell’s law, and explain dispersion.
- Apply total internal reflection to fibre optics, endoscopes and prisms.
- Locate images formed by plane and spherical mirrors, using equations and ray diagrams.
- Use the lensmaker’s equation, the thin-lens equation, magnification and lens power in dioptries.
- Analyse combinations of lenses and the main optical instruments (magnifier, microscope, telescope).
- Explain the optics of the eye, common refractive errors, and how they are corrected.

THINK FIRST

Open your eyes underwater in a swimming pool and everything is badly blurred. Put on a pair of goggles and you see sharply again, even though the goggles' flat plastic windows have no focusing power at all. Why?

Most of your eye's focusing power, about two-thirds of it, comes from refraction at the curved front surface of the **cornea**, where light passes from air ($n = 1.00$) into the cornea ($n \approx 1.38$). Underwater, the outside medium has $n = 1.33$, almost the same as the cornea, so that surface barely bends light, and the eye loses roughly 40 dioptres of focusing power (Problem P22.14). Goggles trap a layer of **air** in front of the cornea, which restores the air–cornea interface and with it the eye's normal focusing.

§22.1

The nature of light and the refractive index

Light is an electromagnetic wave (Chapter 21). In vacuum it travels at $c = 3.00 \times 10^8$ m/s. In a transparent material it travels more slowly, at $v = c/n$, where

$$n = \frac{c}{v}$$

is the **refractive index**. When light passes into a different medium its frequency stays the same, so its wavelength changes: $\lambda_n = \lambda_0/n$.

MATERIAL	n (AT 589 NM)
Vacuum	1 (exactly)
Air	1.0003
Water	1.333
Aqueous and vitreous humour	1.336
Cornea	1.376
Crystalline lens of the eye	1.39–1.41

MATERIAL	n (AT 589 NM)
Crown glass	1.52
Flint glass	1.6–1.7
Diamond	2.42

§22.2

Reflection and refraction

When a ray strikes a smooth boundary, part of it is reflected and part transmitted (refracted). All angles are measured from the **normal** to the surface.

KEY IDEA

Law of reflection: $\theta_r = \theta_i$. The incident ray, the reflected ray and the normal all lie in the same plane.

Snell's law of refraction:

$$n_1 \sin \theta_1 = n_2 \sin \theta_2.$$

Light bends **towards** the normal when it enters a medium of higher index, where it travels more slowly, and **away** from the normal when it enters a medium of lower index.

Both laws follow from **Fermat's principle**: light travels between two points along the path that takes the least time. A lifeguard running along the beach and then swimming to a drowning swimmer faces the same problem. The quickest route bends at the water's edge according to Snell's law.

Dispersion. The refractive index depends slightly on wavelength. In glass, violet light has a higher index than red, so it bends more. Prisms therefore split white light into a spectrum, and raindrops produce rainbows. In lenses, the same effect causes **chromatic aberration**: coloured fringes, because different colours focus at slightly different points.

WORKED EXAMPLE 22.1

Light entering water

A ray in air strikes a water surface at 45° to the normal. Find the angle of refraction, and the speed and wavelength in the water of light with a vacuum wavelength of 589 nm.

Solution

$$\sin \theta_2 = \frac{1.00 \sin 45^\circ}{1.333} = 0.530 \Rightarrow \theta_2 = 32.0^\circ.$$

$$v = \frac{c}{n} = 2.25 \times 10^8 \text{ m/s}, \quad \lambda = \frac{589}{1.333} = 442 \text{ nm}.$$

The light does not change colour as it enters the water. Colour is set by the frequency, which is unchanged.

Apparent depth. Looking straight down into water of depth d , the bottom appears to be at a depth d/n . Rays from the bottom bend away from the normal as they leave the water, so they seem to come from a shallower point. A 2.0 m deep pool looks only 1.5 m deep, a real hazard for people diving in. The same effect makes a straw standing in a glass of water look bent.

§22.3

Total internal reflection

When light goes from a denser medium to a less dense one ($n_1 > n_2$), the refracted ray bends away from the normal. At the **critical angle** θ_c , the refracted ray just skims along the surface ($\theta_2 = 90^\circ$):

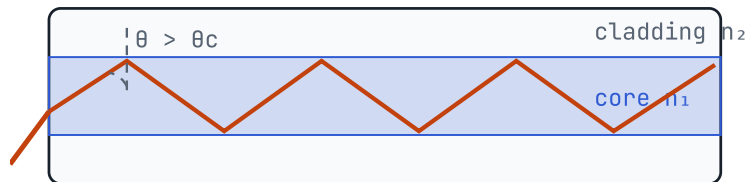
$$\sin \theta_c = \frac{n_2}{n_1}.$$

For any angle of incidence larger than θ_c , there is no refracted ray at all, and **all** the light is reflected. This is **total internal reflection** (TIR). It is more efficient than any mirror coating, reflecting essentially 100% of the light.

- Glass to air ($n = 1.50$): $\theta_c = 41.8^\circ$. Right-angle prisms in binoculars and periscopes use TIR in place of mirrors.
- Diamond to air ($n = 2.42$): $\theta_c = 24.4^\circ$. Light entering a well-cut diamond is trapped by repeated TIR, which is why diamonds sparkle.

Optical fibres

An **optical fibre** has a core of glass with a higher refractive index n_1 , surrounded by a cladding of lower index n_2 . Light entering the core at a shallow enough angle strikes the core–cladding boundary beyond the critical angle, and is guided along the fibre by repeated total internal reflection, even around bends.



light trapped by repeated total internal reflection

An optical fibre. Light striking the core–cladding boundary at more than the critical angle is totally reflected, so it stays trapped in the core as it travels along the fibre.

The largest angle of acceptance at the entrance face (measured in air) is given by the **numerical aperture**:

$$NA = \sin \theta_{\max} = \sqrt{n_1^2 - n_2^2}.$$

WORKED EXAMPLE 22.2

A fibre for an endoscope

A fibre has a core of index 1.62 and a cladding of index 1.52. Find the critical angle at the core–cladding boundary, the numerical aperture, and the maximum acceptance angle in air.

Solution

$$\theta_c = \sin^{-1} \frac{1.52}{1.62} = 69.8^\circ, \quad NA = \sqrt{1.62^2 - 1.52^2} = \sqrt{0.314} = 0.56, \quad \theta_{\max} = \sin^{-1} 0$$

Evaluate. A wide acceptance angle helps a fibre collect light efficiently, which suits medical endoscopes. Communication fibres use cores and claddings with very similar indices, so they have a small NA. This keeps the many possible ray paths nearly equal in length, so short pulses do not smear out.

IN PRACTICE

Fibre optics in medicine. An **endoscope** (a gastroscope, colonoscope, bronchoscope or arthroscope) carries light into the body along a bundle of illumination fibres. A *coherent* bundle of tens of thousands of fibres, arranged identically at both ends, or a tiny camera chip at the tip, brings the image back. This lets doctors inspect and operate inside the body through natural openings or keyhole incisions. Fibres also deliver laser energy for surgery, for example to break up kidney stones or seal blood vessels, and they carry light to and from sensors such as pulse oximeters and blood-oxygen probes. Optical-fibre networks carry almost all the world's internet traffic.

§22.4

Plane and spherical mirrors

Plane mirrors

A plane mirror forms a **virtual** image as far behind the mirror as the object is in front of it. The image is upright and the same size, but reversed front-to-back, which is why your mirror image appears to swap left and right.

Spherical mirrors

A **concave** mirror (curving towards the object) brings parallel rays to a real focal point. A **convex** mirror makes them diverge, as if from a virtual focal point behind it. For rays close to the axis (*paraxial* rays), the focal length is half the radius of curvature:

$$f = \frac{R}{2}.$$

KEY IDEA

Mirror and lens equation, with magnification:

$$\frac{1}{s} + \frac{1}{s'} = \frac{1}{f}, \quad m = \frac{h'}{h} = -\frac{s'}{s}.$$

Sign convention (“real is positive”):

- s is positive for a real object, which is the usual case.
- s' is positive for a **real** image (in front of a mirror, or behind a lens) and negative for a **virtual** image.
- f is positive for converging elements (concave mirrors and convex lenses) and negative for diverging ones.
- $m > 0$ means the image is upright, and $m < 0$ means it is inverted.

NOTE

H. C. Verma and many school texts use the **Cartesian** sign convention. There, all distances are measured from the mirror or lens, positive in the direction the incident light travels, and the lens equation is written $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$. The two conventions give the same physical answers. Pick one and use it consistently.

Principal rays for drawing ray diagrams, for a mirror or a lens:

1. A ray parallel to the axis passes (or appears to pass) through the focal point.
2. A ray through the focal point emerges parallel to the axis.
3. For a mirror, a ray through the centre of curvature returns along itself. For a lens, a ray through the centre of the lens passes straight through, undeviated.

WORKED EXAMPLE 22.3**A concave make-up mirror**

A concave mirror has a focal length of 10 cm. Find the image position, the magnification and the nature of the image for an object at (a) 30 cm and (b) 5.0 cm.

Solution

(a) Object at 30 cm:

$$\frac{1}{s'} = \frac{1}{10} - \frac{1}{30} = \frac{2}{30} \Rightarrow s' = 15 \text{ cm}, \quad m = -\frac{15}{30} = -0.50.$$

The image is real, inverted and half-size, 15 cm in front of the mirror.

(b) Object at 5.0 cm:

$$\frac{1}{s'} = \frac{1}{10} - \frac{1}{5.0} = -\frac{1}{10} \Rightarrow s' = -10 \text{ cm}, \quad m = -\frac{-10}{5.0} = +2.0.$$

The image is virtual, upright and twice the size, 10 cm *behind* the mirror.

Evaluate. Make-up and shaving mirrors, and the small mirrors dentists use, are concave mirrors used with the face *inside* the focal length, to give a magnified, upright image.

WORKED EXAMPLE 22.4

A convex security mirror

A convex mirror with $f = -20$ cm shows a person standing 2.0 m away. Find the image position and magnification.

Solution

$$\frac{1}{s'} = -\frac{1}{20} - \frac{1}{200} = -\frac{11}{200} \Rightarrow s' = -18 \text{ cm}, \quad m = -\frac{-18.2}{200} = +0.091.$$

The image is virtual, upright and small. Convex mirrors give a wide field of view, which is why they are used in shops and on car wing mirrors. The small image makes objects seem farther away than they are, hence the warning printed on the mirror: “objects in mirror are closer than they appear”.

§22.5

Refraction at a spherical surface and thin lenses

At a single spherical surface of radius R separating media of index n_1 (where the light starts) and n_2 :

$$\frac{n_1}{s} + \frac{n_2}{s'} = \frac{n_2 - n_1}{R}.$$

Here R is positive if the centre of curvature is on the outgoing side of the surface. The quantity

$(n_2 - n_1)/R$ is the **power** of the surface. Apparent depth is the special case of a flat surface, $R \rightarrow \infty$.

A **thin lens** has two such surfaces close together. Applying the surface equation twice gives the **lensmaker's equation**:

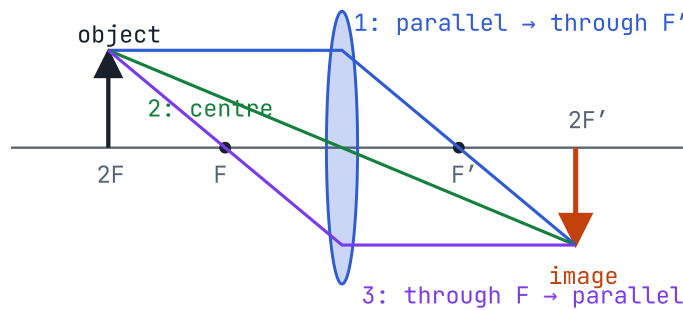
$$\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right),$$

for a lens of index n in air, with the same sign rule for each radius. Images then obey the same equation as mirrors:

$$\frac{1}{s} + \frac{1}{s'} = \frac{1}{f}, \quad m = -\frac{s'}{s}.$$

- A **converging** lens, thicker in the middle, has $f > 0$.
- A **diverging** lens, thinner in the middle, has $f < 0$.

Lens power is $P = 1/f$, measured in **dioptries** (D, which is m^{-1}). Opticians prescribe in dioptries because, for thin lenses in contact, powers simply **add**: $P = P_1 + P_2$.



Ray diagram for a converging lens with the object at $2F$. The three principal rays meet at a real, inverted, same-size image at $2F'$.

WORKED EXAMPLE 22.5

Designing and using a lens

(a) A symmetric biconvex lens with $n = 1.50$ has surfaces of radius 20 cm. Find its focal length and power. (b) An object is placed 30 cm in front of it. Find the image position and magnification. (c) Two thin lenses, with $f_1 = +20$ cm and $f_2 = -30$ cm, are placed in contact. Find the combined focal length.

Solution

(a) Here $R_1 = +20$ cm and $R_2 = -20$ cm:

$$\frac{1}{f} = 0.50 \left(\frac{1}{20} + \frac{1}{20} \right) = \frac{1}{20} \Rightarrow f = 20 \text{ cm}, \quad P = 5.0 \text{ D}.$$

(b) Object at 30 cm:

$$\frac{1}{s'} = \frac{1}{20} - \frac{1}{30} = \frac{1}{60} \Rightarrow s' = 60 \text{ cm}, \quad m = -2.0.$$

The image is real, inverted and twice the size. A slide projector works this way, with the slide just outside the focal length.

(c) Lenses in contact:

$$P = \frac{1}{0.20} + \frac{1}{-0.30} = 5.0 - 3.3 = 1.7 \text{ D} \Rightarrow f = 60 \text{ cm}.$$

Combining a converging lens with a weaker diverging lens of a different glass is how *achromatic doublets* cancel chromatic aberration.

§22.6

The eye

The eye is a remarkable optical instrument. Light is refracted mainly at the **cornea**, which provides about 43 D of power, and then by the **crystalline lens**, which provides about 15–20 D, and is focused onto the **retina**, about 17 mm behind the lens. The total power of a relaxed eye is about 60 D. The iris adjusts the size of the pupil, from about 2 mm to 8 mm. The lens is flexible. To focus on near objects, the ciliary muscle contracts, the lens bulges, and its power increases. This is **accommodation**.

- **Far point:** the farthest distance the eye can focus on. For a normal eye it is infinity.
- **Near point:** the closest distance the eye can focus on. Conventionally it is taken as 25 cm for a young adult. It recedes with age, because the lens stiffens (presbyopia).

Refractive errors and their correction

CONDITION	PROBLEM	CORRECTION
Myopia (short sight)	The eye is too powerful or too long, so distant objects focus in front of the retina. The far point is closer than infinity.	A diverging lens that makes distant objects appear to lie at the far point: $P = -1/d_{\text{far}}$.
Hyperopia (long sight)	The eye is too weak or too short. The near point is farther than 25 cm.	A converging lens that makes an object at 25 cm appear to lie at the near point.
Presbyopia	Loss of accommodation with age.	Reading glasses or bifocals (converging).
Astigmatism	The cornea is curved differently in different directions.	A cylindrical lens.

WORKED EXAMPLE 22.6

Prescribing glasses

(a) A myopic patient's far point is 50 cm. What lens power corrects their distance vision? (Ignore the small gap between the lens and the eye.) (b) A hyperopic patient's near point is 1.0 m. What reading lens lets them read at 25 cm?

Solution

(a) The lens must form a virtual image of a distant object ($s = \infty$) at the far point, so $s' = -0.50$ m:

$$P = \frac{1}{s} + \frac{1}{s'} = 0 - \frac{1}{0.50} = -2.0 \text{ D.}$$

(b) An object at $s = 0.25$ m must give a virtual image at the near point, $s' = -1.0$ m:

$$P = \frac{1}{0.25} + \frac{1}{-1.0} = 4.0 - 1.0 = +3.0 \text{ D.}$$

IN PRACTICE

Refractive surgery and lens implants. LASIK reshapes the cornea with an excimer laser. For myopia, flattening the central cornea reduces its power, and a change in radius of only about 0.4 mm corrects -2.5 D (Problem P22.18). In **cataract surgery**, the clouded natural lens is replaced by an artificial *intraocular lens*, whose power is calculated from the length of the eye and the curvature of the cornea. This is the most common surgical operation in the world. **Ophthalmoscopes** and **retinal cameras** use the eye's own optics, in reverse, to image the retina.

§22.7

Optical instruments

The simple magnifier. We judge size by the angle an object subtends at the eye. Bringing an object closer increases that angle, but only down to the near point, $N = 25$ cm. A converging lens of focal length f lets the object come closer while the eye still sees a clear (virtual) image. The angular magnification is

$$M = \frac{N}{f} \quad (\text{image at infinity, relaxed eye}), \quad M = 1 + \frac{N}{f} \quad (\text{image at the near point}).$$

The compound microscope. An objective lens of short focal length f_o forms a magnified real image, and an eyepiece of focal length f_e acts as a magnifier for that image. With a tube length L (the distance between the focal points of the two lenses), the total magnification is

$$M \approx -\frac{L}{f_o} \cdot \frac{N}{f_e}.$$

The refracting telescope. The objective forms an image of a distant object at its focal point, and the eyepiece magnifies it:

$$M = -\frac{f_o}{f_e}.$$

The large objective (or mirror, in a reflecting telescope) also collects more light, and it gives better resolution, which is limited by diffraction (Chapter 23).

WORKED EXAMPLE 22.7

A laboratory microscope

A microscope has an objective with $f_o = 4.0$ mm, an eyepiece with $f_e = 25$ mm, and a tube length of 160 mm. Find its magnification.

Solution

$$M = -\frac{160}{4.0} \times \frac{250}{25} = -40 \times 10 = -400.$$

The objective magnifies 40× and the eyepiece 10×, giving a total of 400×. The image is inverted. As Chapter 23 shows, diffraction limits useful light-microscope magnification to about 1000–1500×, however the lenses are combined.

CHAPTER SUMMARY

- $n = c/v$, and $\lambda_n = \lambda_0/n$, with the frequency unchanged. Reflection: $\theta_r = \theta_i$. Snell: $n_1 \sin \theta_1 = n_2 \sin \theta_2$.
- Total internal reflection happens when $n_1 > n_2$ and $\theta > \theta_c$, with $\sin \theta_c = n_2/n_1$. Fibres: $\text{NA} = \sqrt{n_1^2 - n_2^2}$.
- Mirrors: $f = R/2$. Mirrors and thin lenses: $1/s + 1/s' = 1/f$ and $m = -s'/s$, in the real-is-positive convention.
- Spherical surface: $n_1/s + n_2/s' = (n_2 - n_1)/R$. Lensmaker: $1/f = (n - 1)(1/R_1 - 1/R_2)$. Power $P = 1/f$ in dioptres, and powers add for lenses in contact.
- Eye: about 60 D in total, two-thirds of it from the cornea. Myopia: $P = -1/d_{\text{far}}$. Hyperopia: $P = 4 - 1/d_{\text{near}}$ (distances in metres).
- Magnifier: $M = N/f$. Microscope: $M \approx -(L/f_o)(N/f_e)$. Telescope: $M = -f_o/f_e$.

Practice problems

Full step-by-step solutions are in the separate solutions PDF.

LEVEL A – CONCEPT CHECK

P22.1

Why does a straw standing in a glass of water appear bent at the water surface? Draw a ray diagram to explain.

P22.2

Explain why a well-cut diamond sparkles much more than a glass imitation of the same shape.

P22.3

Can a convex mirror ever form a real image of a real object? Explain using the mirror equation.

P22.4

Explain why your vision is badly blurred underwater, but clear again when you wear swimming goggles.

LEVEL B – STANDARD PROBLEMS

P22.5

A ray strikes a 5.0 cm thick glass slab ($n = 1.50$) at 60° to the normal. Find the angle of refraction inside the slab, the direction of the ray when it emerges, and the sideways displacement of the emerging ray.

P22.6

An optical fibre has a core of index 1.62 and a cladding of index 1.52 (as in Example 22.2). If the fibre were used without its cladding, in air, what would its numerical aperture be? Why is the cladding nevertheless essential?

P22.7

A man wants a shaving mirror that gives him an upright image of his face magnified twice when his face is 15 cm from the mirror. What kind of mirror, and what radius of curvature, does he need?

P22.8

A car's convex wing mirror has a radius of curvature of 1.0 m. A following car is 10 m behind it. Find the image position and magnification.

P22.9

A converging lens has a focal length of 10 cm. Find the image position, magnification and nature of the image for an object at (a) 15 cm and (b) 5.0 cm. Sketch a ray diagram for each.

P22.10

A thin converging lens with $f = 20$ cm and a thin diverging lens with $f = -30$ cm are placed in contact. Find the focal length and power of the combination. Is it converging or diverging?

P22.11

A 50-year-old's near point has receded to 75 cm. What reading glasses (power in dioptries) will let them read comfortably at 25 cm?

P22.12

A magnifying glass has a focal length of 5.0 cm. Find its angular magnification when the image is (a) at infinity and (b) at the near point (25 cm).

LEVEL C – CHALLENGE PROBLEMS**P22.13**

The fish in the bowl. A goldfish swims 10 cm from the front of a spherical bowl of radius 15 cm, filled with water ($n = 1.33$). Ignore the thin glass. Using the equation for refraction at a spherical surface, find where the fish appears to be to an observer looking in from the front, and the magnification of its image.

P22.14

The cornea underwater. The front surface of the cornea has a radius of curvature of 7.8 mm, and the cornea has an index of 1.376. (a) Find the power of this surface in air. (b) Find its power when the eye is underwater ($n = 1.333$). (c) How many dioptries of focusing power are lost? Explain why goggles restore clear vision.

P22.15

A microscope has an objective with $f_o = 1.6$ cm and an eyepiece with $f_e = 2.5$ cm, separated by 22.1 cm. The final image is at infinity. (a) Where is the intermediate image? (b) Where must the object be placed? (c) Find the overall magnification, taking the near point as 25 cm.

P22.16

Prisms and dispersion. A glass prism has an apex angle of 60° . (a) At minimum deviation, the ray passes symmetrically through the prism, and $n = \frac{\sin \frac{1}{2}(A + \delta_{\min})}{\sin \frac{1}{2}A}$. Find n if $\delta_{\min} = 37.2^\circ$. (b) For this glass, $n_{\text{red}} = 1.513$ and $n_{\text{violet}} = 1.532$. Find the minimum-deviation angles for red and violet light, and the angular spread between them.

P22.17

An object is placed 30 cm in front of a converging lens with $f_1 = 10$ cm. A second converging lens, with $f_2 = 15$ cm, is 25 cm beyond the first. Find the position, magnification and nature of the final image.

P22.18

LASIK. Model the eye's front corneal surface as a single refracting surface from air ($n = 1$) into the cornea ($n = 1.376$), with a radius of 7.80 mm. (a) Find its power. (b) A patient's eye has 2.5 D too much power (myopia with a far point of 40 cm). To what radius must the central cornea be reshaped to correct this? (c) By how many micrometres must the radius change? Comment on why the surgery needs such precise laser control.

All worked examples and practice problems in these notes are original.
Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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